

A Classification and Recognition Method for Motor Imagery EEG Signals using Fusion of DRSN and WPD

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Abstract—Noise reduction is a key factor in determining the effectiveness of motor imagery (MI) EEG signal classification and recognition. Deep Residual Shrinkage Network (DRSN) is suitable for denoising MI EEG signals due to its soft thresholding and attention mechanism characteristics. However, there is a lack of intermediate scale features in its processing, which are crucial for examining functional connectivity between brain regions. Therefore, this article proposes a multi-scale feature learning method that combines DRSN and Wavelet Packet Decomposition (WPD) to enhance DRSN's ability to capture intermediate scale features. Through WPD, EEG signals are decomposed into different frequency bands, covering from δ wave to γ wave. The different frequency ranges of waves enable the network to effectively capture intermediate scale signal features of dynamic changes such as event related potentials, EEG synchronization and desynchronization. The experimental results show that the average accuracy of DRSN on the BCI competition IV 2a and 2b datasets is increased to 84.45% and 91.26% after the introduction of WPD, which can ensure the effective recognition of MI EEG signals.

Keywords—EEG signals, motor imagery, deep residual shrinkage network, wavelet packet decomposition, ERS, ERD

I. INTRODUCTION

With the popularization of brain computer interface technology, more and more attention has been paid to the devices and technologies that the brain interacts with the external environment. For these devices and technologies, it is crucial to accurately classify and identify EEG signals[1]. At present, the acquisition of MI EEG signals mainly depends on non-invasive electrodes and head-worn EEG acquisition equipment. Compared with the invasive EEG acquisition, the disadvantage of this method is that the acquired signal is weak and the noise level is large. In addition, the interference of physiological artifacts (such as eye electricity and myoelectricity) and the physiological differences between different individuals further increase the diversity of noise in the signal, making the traditional signal denoising methods have limitations here[2], so it is necessary to find a more effective denoising recognition method for MI EEG signals.

In recent years, many MI EEG signal classification methods based on traditional machine learning have achieved remarkable results. Support vector machine (SVM), as a widely used classifier, is often used for EEG signal classification. SVM performs well for processing EEG data with high-dimensional features. For example, Teng et al.[3] used SVM combined with KNN algorithm to propose a hybrid model for EEG emotion classification, successfully improving the classification accuracy. Random Forest (RF) is also used

for EEG signal classification due to its ability to handle multidimensional data and its good noise immunity[4]. For example, Zhou et al.[5] used RF to classify MI EEG signals and achieved high accuracy. In addition, Naive Bayes and K-Nearest Neighbors (KNN) are also applied in EEG classification. Naive Bayes uses conditional probability to achieve category prediction, which is suitable for processing noisy data, while KNN classifies by calculating the distance between data points, which is suitable for processing nonlinear data[6-7].

In addition, deep learning methods have also made great progress in the classification of EEG signals. Annu Kumari et al.[8] proposed a method for channel selection and classification of MI EEG signals based on hybrid optimization and two-level deep learning. This method effectively reduces redundant information in EEG signals by combining genetic algorithms and deep learning, improving classification accuracy. Ruilin Li et al.[9] developed "TFormer", a Transformer model based on time-frequency transformation, which is applied to driver fatigue detection. The model effectively captures the time-frequency characteristics of EEG signals by combining batch normalization and Transformer structure, demonstrating excellent performance in complex tasks. In addition, R. Li et al.[10] proposed a sample enhancement technique based on the intrinsic characteristics of EEG, which enhances the generalization ability of the model by data augmentation of the original EEG signal. This method provides an effective approach to solve the problem of EEG signal classification with limited training samples[11-13].

DRSN effectively overcomes the limitation of setting fixed threshold in traditional noise reduction methods. Using soft thresholding and attention mechanism, it has been widely used in the classification and recognition of high noise signals, and has achieved high accuracy. In terms of speech emotion recognition, DRSN's classification accuracy of different emotions is 89.96%[14]; When processing MI EEG signals, which have significant temporal characteristics and long-term temporal dependence, DRSN also shows better performance than traditional noise reduction recognition methods. DRSN has significantly improved the accuracy of four types of MI action recognition in dataset 2a, which are 77.82% for the left hand, 77.35% for the right hand, 77.98% for both feet, and 75.96% for the tongue[15-17]. However, the requirements of application scenarios such as brain disease recognition and diagnosis are still not satisfied, that is, DRSN still has some limitations in the classification and recognition of MI EEG signals. This is because EEG signal, as a signal with significant temporal characteristics, is characterized by the frequency band baud characteristics generated by different

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This work was supported by the National Natural Science Foundation of China under Grant No. 61806118.

brain area activities throughout the whole signal[18-21]. However, the convolution module structure within DRSN makes it focus on local and global scale features in the process of processing MI EEG signals, while the capture of signal mid-scale features is missing. This limitation leads to poor performance of DRSN in processing multi-scale, high-dimensional EEG signal classification and recognition tasks.

WPD is an extension of the traditional wavelet transform, which allows more detailed multi-level decomposition of signals to form multiple sub bands with different frequencies[22]. These sub bands reveal the complex frequency information of signals at multiple scales, which is particularly critical for EEG waveform analysis. In MI EEG signal processing, WPD is suitable for extracting key features such as event related potentials, rhythmic changes of EEG spectra, and synchronization and desynchronization of EEG. In this way, WPD can accurately capture the intermediate features of EEG activity, so as to achieve a richer and more accurate feature representation of EEG signals, which can greatly improve the performance of DRSN in classifying and detecting MI EEG signals.

Therefore, this paper uses WPD to increase the multi-scale receptive field of DRSN model and further improve its ability to learn the characteristics of MI EEG signals. In the part of MI EEG signal preprocessing, WPD is used to subdivide the EEG signal into different frequency bands, and then α wave and β wave, the two signal bands most related to MI, are selected to extract features by calculating the average power change. After adjusting the stack number of basic residual shrinkage blocks in DRSN, the feature vector is input into the network for signal recognition and classification. Finally, the network model after WPD fusion is tested on BCI-IV-2a and 2b datasets, and compared with the network architecture without WPD fusion and other commonly used models and classifiers for recognition performance, so as to improve the performance of DRSN in MI EEG signal detection and classification.

II. DRSN MODEL INTEGRATING WPD

Research shows that MI involves prefrontal and parietal regions of the brain, which play a key role in processing motor planning and execution. The two frequency bands of alpha wave (8-13Hz) and beta wave (13-30Hz) in EEG signals are closely related to the activity of brain regions, and can reflect the dynamic changes of the above regions in MI[23]. For the four types of MI tasks in dataset 2a, the changes of alpha and beta waves are particularly significant. In the MI tasks of hands and feet, the amplitude of alpha and beta waves decreases. This change is called Event Related Desynchronization (ERD). ERD reflects the increase of activities in specific brain regions during exercise preparation and imagination. In addition, the rebound of beta wave after task execution is more obvious in hand tasks, that is, Event Related Synchronization (ERS), which indicates the process of cortical reorganization after motion execution. The above ERD and ERS patterns are the intermediate scale features that need extra attention in the classification and recognition of MI EEG signals.

A. Overview of wavelet packet decomposition

Because DRSN is missing in processing the intermediate scale features of MI EEG signals, this paper plans to use WPD combined with DRSN to increase the network's ability to

capture the intermediate scale features of MI EEG signals, thus increasing the network's ability to perceive multi-scale features. WPD is an extension of wavelet transform, which considers not only the low-frequency part of the signal concerned in wavelet transform, but also the high-frequency part. At each decomposition level, not only low-frequency signals are further decomposed, but also high-frequency signals are decomposed, thus more detailed frequency information can be obtained. The choice of wavelet basis function determines the result of wavelet packet decomposition, which has a certain impact on the feature extraction and classification of MI signals. There are many wavelet bases to choose from in wavelet packets. Haar wavelet basis is applicable to the situation where there are many jumps in the signal, while trigonometric function basis is applicable to the signal containing sine component or high-frequency oscillation[24]. Since EEG data are discrete data collected by external equipment, wavelet basis suitable for discrete transformation should be selected when WPD is performed, and Daubechies (dbN) wavelet basis is often used. In this paper, db4 wavelet basis is selected to decompose MI EEG signal. The tree hierarchy diagram is shown in Figure 1.

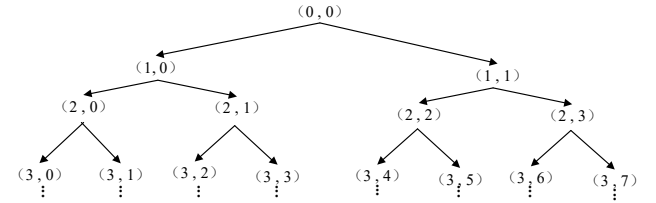


Fig. 1. Schematic diagram of the WPD tree hierarchical structure of MI EEG signals

Secondly, choosing the appropriate number of decomposition layers is the key to effectively use WPD for signal decomposition. According to the characteristics of wavelet packet decomposition, this can be expressed as:

$$lay = lb(Sw / f_{de}) \quad (1)$$

Where, lay is the final number of layers to be decomposed, Sw is the signal bandwidth, and f_{de} is the bandwidth. The sampling frequency of BCI data set is 250Hz, and the f_{de} value is 8Hz. If you bring it into the Equation, you can see that four layer decomposition is required.

As shown in Figure 1, the first line (0,0) represents the preprocessed EEG signal, and the subsequent lines represent the decomposition of each layer. Through the first level decomposition, the signal frequency band is divided into two parts to generate two sub bands containing different frequencies. By analogy, wavelet packet decomposition can be expressed as:

$$d_l^{j,2n} = \sum_k h_{k-2l} d_k^{j-1,n} \quad j, k \in Z, n \in Z_+ \quad (2)$$

$$d_l^{j,2n+1} = \sum_k g_{k-2l} d_k^{j-1,n} \quad j, k \in Z, n \in Z_+ \quad (3)$$

Where, d is the frequency band coefficient of each layer, j represents the number of layers, k represents the nodes of each layer, and n represents the frequency. According to Equation (2) and Equation (3), WPD actually decomposes the signal into different frequency bands through a combination of high and low pass filters[25-26].

The original EEG signal is processed into four layers of WPD through the above process. Because C3 and C4 channels are located near the motor cortex of the brain, corresponding

to the motor areas of the left and right hemispheres; Cz is located in the central sulcus of the brain, reflecting the activity of bilateral motor areas[27], so three signal channels, C3, C4, and Cz, are selected to extract the node coefficients of three relevant sub bands covering the α and β frequency bands (8-30Hz) in the fourth layer and reconstruct the signal to obtain the reconstructed signal. Perform feature extraction on the following steps:

Step 1: Select baseline time window 2.5-3s and task time window 3-7s.

Step 2: extracts the power spectral density of the baseline period and the mission period respectively.

Step 3: Calculate the average power in the baseline period and the mission period respectively for the time window and frequency range:

$$\bar{P} = \frac{1}{N_t N_f} \sum_{t=t_1}^{t_2} \sum_{f=f_1}^{f_2} P(f, t) \quad (4)$$

Where, N_t represents the number of time points in the time window, and N_f represents the number of frequency points in the frequency range.

Step 4: Calculate the power change from the baseline period to the task period:

$$\Delta P = (\bar{P}_{baseline} - \bar{P}_{task}) / \bar{P}_{baseline} \quad (5)$$

Where, $\bar{P}_{baseline}$ represents the average power in the baseline period, and $\bar{P}_{task}$ represents the average power in the mission area.

After the above steps and in combination with the time-frequency distribution diagram, the power change can be visualized, as shown in Figure 2.

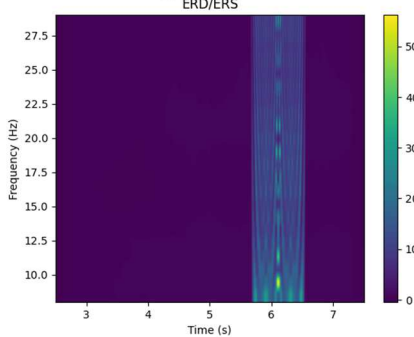


Fig. 2. Reconstruct the time-frequency distribution map of power changes in signal S_{re}

It can be seen from Figure 2 that the time-frequency distribution has a large change before and after the 6s time point, and there is a significant increase in power in the time range from 5.7s to 6.2s, indicating that the event synchronization (ERD) within the frequency range at this time is enhanced, and the subjects are in motion imagination. For a period of time after 6.2s, the power gradually decreases and the change is not significant, indicating that the subject's motion imagination is over at this time, which is the post event synchronization (ERS). Therefore, processing MI EEG signals through WPD can make the mesoscale characteristics of ERD and ERS more obvious.

To sum up, after WPD of MI EEG signals, feature extraction can obtain a feature array containing the average power changes of all samples in C3, C4, and Cz channels, and its dimension is (288,3,3). The ERD/ERS mid-scale features in this array are more prominent, which can be input into

DRSN as the overall features of MI EEG signals for subsequent classification and recognition.

B. DRSN model architecture

The introduction of residual shrinkage module makes DRSN perform better in MI EEG signal classification and recognition. In this paper, DRSN is mainly composed of two parts: basic residual shrinkage module (BasicBlock) and residual shrinkage network (RSNet)[28]. Among them, BasicBlock is the basic unit of the network and the core component of DRSN. In addition to the basic convolution layer, it also contains the Shrinkage module, which can be used for threshold learning and feature shrinkage, and adaptively generate feature thresholds to reduce noise and redundancy. The model structure diagram of BasicBlock is shown in Figure 3.

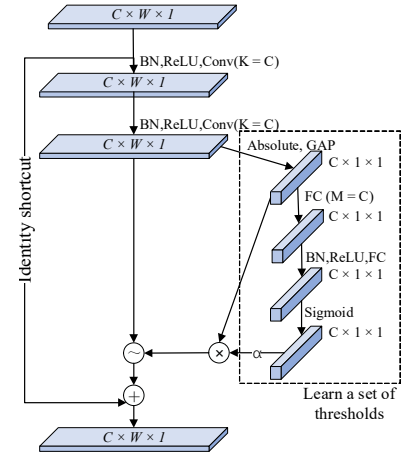


Fig. 3. BasicBlock module composition

The right box in Figure 3 is the Shrinkage module. As a threshold learning subnet, this module is mainly responsible for further noise reduction and shrinkage processing of input features. It can adaptively set different thresholds for different input features to achieve noise reduction, that is, soft thresholding shrinkage. In this process, the part of the signal that is higher than the threshold value is reserved, and the part that is lower than the threshold value is set to 0, as shown in Equation (6).

$$S(x, \lambda) = \text{sign}(x) \max(|x| - \lambda, 0) \quad (6)$$

Where, x is the input signal, λ is the threshold value, and $\text{sign}(x)$ is the sign function of the signal. Combined with Equation (6), we can see that the difference between the input signal and the threshold value is obtained by subtracting the input signal from the threshold value. If the signal is greater than the threshold value, the difference value is positive, otherwise it is negative. The maximum value of the difference value and 0 will be the signal lower than the threshold value for noise suppression, while retaining the signal higher than the threshold value. Finally, the original signal symbols are retained to complete the signal noise reduction process, and then the subsequent feature extraction is performed in BasicBlock.

The BasicBlock module uses 3x3 convolution to check the input MI EEG signal for processing, and uses BatchNorm2d batch normalization and ReLU activation function for further processing, and then uses Shrinkage module to reduce noise and feature shrinkage. At the end of the module, 1x1 convolution performs dimension matching and residual connection to enhance feature representation. The DRSN model as a whole is composed of multiple BasicBlock

modules stacked in layers. Each layer is responsible for processing features of different scales. At the same time, the input features can be down sampled to reduce the space size of the feature map and increase its depth. Of course, as the stack level increases, the model depth and convergence time gradually increase, and the model may be over fitted. Therefore, selecting an appropriate BasicBlock module stack level has a great impact on the recognition and classification performance of DRSN. At the end of the network, MI EEG signals are finally classified through global average pooling and full connection layer. The network structure of DRSN is shown in Figure 4.

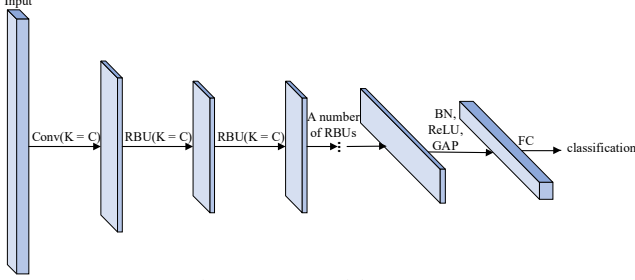


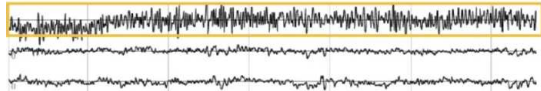
Fig. 4. DRSN model structure

C. Learning of threshold λ

The quality of MI EEG signal is affected by two main kinds of noise. Its generation mechanism and characteristics are shown in Table I[29][30], and the noise waveform is shown in Figure 5.

TABLE I. THE MECHANISM AND CHARACTERISTICS OF TWO MAIN TYPES OF NOISE IN MI EEG SIGNALS

Noise Source	Physiological Artifact	Non-invasive EEG Acquisition Equipment
Generation mechanism	Natural physiological activities (such as blinking, heartbeat, etc.) produce electrical signals that overlap the frequency range of EEG signals, resulting in interference in EEG signals.	The noise generated by poor contact between electrodes and skin, environmental electromagnetic interference, electronic noise of equipment itself and other factors is superimposed on EEG signal during signal acquisition and transmission.
characteristic	It has large amplitude and specific pattern, and individual difference is large.	It has a relatively random characteristic, which is manifested as high-frequency noise or uniformly distributed interference in the whole frequency band.



a Power frequency interference waveform

b Blink artifact waveform

Fig. 5. Noise waveform schematic diagram

The soft thresholding process in DRSN does not depend on the specific type of noise. It eliminates noise by generating dynamic threshold λ to reduce the signal amplitude. According to the description in Table I and Figure 5, the characteristics of the two main types of noise waveforms are relatively obvious. For the physiological artifacts with large values, the signal amplitude can be reduced by soft thresholding for noise reduction. For the random electronic noise in the EEG acquisition equipment, with the 50-60Hz power frequency interference value as the standard, the channel data with more serious noise pollution can be removed by band-pass filtering during preprocessing, and the other channel data with less pollution can be de noised by soft thresholding.

In conclusion, DRSN's process of denoising for EEG signal contraction is based on the accurate learning of threshold λ . The threshold learning process is mainly realized through the Fully Connect (FC) layer inside the Shrinkage module. This layer can adaptively learn the optimal threshold of each feature channel from the input feature data. As the training process progresses, the FC layer gradually generates a threshold for each channel. This threshold can maximize the network's response to signal patterns closely related to MI, while suppressing insignificant features such as noise, background activities, or signal components unrelated to tasks. In the Shrinkage module, after taking the absolute value of the input feature vector of each channel, the dimension is reduced to the average value through global average pooling, and then these features are input into the FC layer, including BatchNorm1d batch normalization and ReLU activation functions. Finally, the output is normalized to between 0 and 1 using the sigmoid activation function. In this way, the Shrinkage module can generate a group of dynamic thresholds to perform soft thresholding shrinkage noise reduction on the input signal. The learning of the adaptive noise reduction threshold λ and the process of soft thresholding the signal according to the threshold λ are shown in Figure 6.

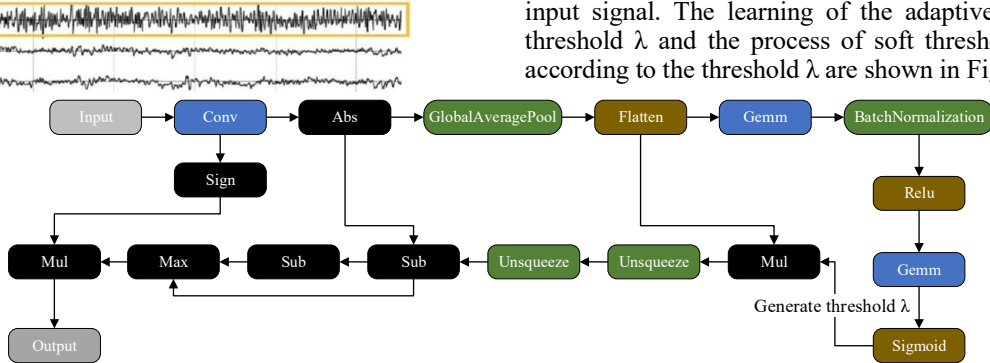


Fig. 6. Shrinkage module adaptive denoising threshold learning process

III. EXPERIMENTAL PROCESS AND RESULT ANALYSIS

A. Data sources

This paper uses BCI competition IV 2a and 2b data sets[30] to verify the feasibility of the proposed method. The sampling

frequency of the data set is 250Hz, and the band-pass filter range is 0.5-100Hz.

Data set 2a contains EEG signals of 9 subjects in MI state, numbered A01~A09, covering four different MI tasks, namely left hand, right hand, foot and tongue, corresponding to

categories 0, 1, 2 and 3. The experimental paradigm is shown in Figure 7 (a). At the beginning of the experiment, participants face the screen, and the screen displays a black background and a fixed cross point, while the buzzer gives a prompt sound. After 2 seconds, four categories of MI task prompts will appear on the screen, each of which lasts for 1.25 seconds. The subjects need to make corresponding MI status according to the icons on the screen[31].

Data set 2b also contains EEG signals of 9 subjects in MI state, numbered B01~B09, covering two different MI tasks, namely left hand and right hand. The experimental paradigm is shown in Figure 7 (b). After the experiment began, a fixed cross point appeared in the center of the screen. The buzzer gave a prompt tone at the second second. After 1-2 seconds, two categories of MI task prompts appeared. The subjects made corresponding MI status according to the prompts. After the end of this MI, they were given a 1.5s rest time, and then entered the next Trail.

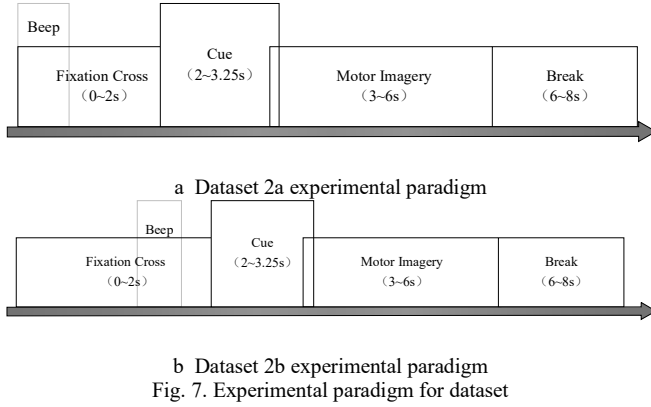


Fig. 7. Experimental paradigm for dataset

B. Data preprocessing

Take dataset 2a as an example to show the preprocessing process: first, read the EEG data file to obtain the original EEG signal, and check the specific information of the EEG data, including the sampling rate, channel number, etc., then obtain the type of event and the mark mark corresponding to the event, which marks ten different events in the MI experiment from generation 1 to 10. This experiment only needs to pay attention to the marks related to left hand, right hand, feet and tongue movements, that is {'769': 7, '770': 8, '771': 9, '772': 10}, corresponding to the above four MI tasks, where '769', '770', '771' and '772' represent four event type IDs respectively. Since the frequency range of 7-35 Hz covers the main frequency bands related to MI tasks in EEG signals, the band-pass filter is used to filter the EEG data at 7-35 Hz. Since eye movements may introduce artifacts unrelated to EEG activities, the three eye electrical channels "EOG left", "EOG central" and "EOG right" data are removed to reduce interference. At present, the number of remaining effective channels is 22.

Because the experiment only needs the data of the MI phase, it can be seen from the introduction of the data set that in a trial, the stimulus event starts from 2s, and the MI phase lasts from 3s to 6s. Therefore, in order to ensure that the analyzed signal is most relevant to the target task, the experiment uses data from 1s to 4s after the event. The subjects in this time range focus on the MI stage, so after setting the minimum and maximum sampling time, the EEG data is segmented epoch in combination with the above four mark marks. Table II shows the list of epoch parameters after segmentation.

TABLE II. EEG SIGNAL SEGMENTATION EPOCH PARAMETER TABLE

Epoch	Parameter
Number of events	288
Events	769: 72 770: 72 771: 72 772: 72
Time range	1.000 – 4.000 sec
Baseline	off

From Table II, we can see that there are 288 events in the EEG data after cutting, and the ID of the four event types is 769 770 771 772, corresponding to 72 times of each event in the data set. The duration of this epoch is from 1s to 4s. Finally, the epochs of the four events are globally averaged to generate an evoked object, which records the average responses of multiple epochs. The MI EEG signal after the above segmentation, global average reference and other processing is shown in Figure 8. Finally, the preprocessed EEG signal data and tags are obtained by obtaining the events and data data of the segmentation epoch. So far, the pre-processing of the original EEG signal is completed, and the pre-processing process is shown in Figure 9.

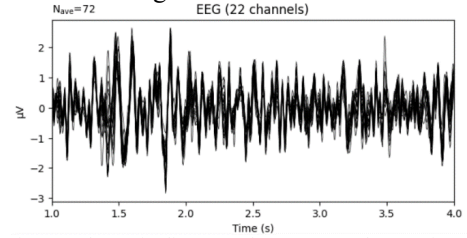


Fig. 8. Schematic diagram of preprocessed EEG signals

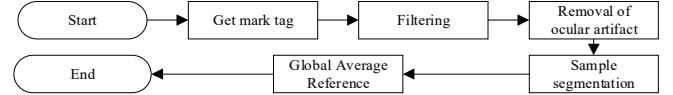


Fig. 9. Preprocessing process of motor imagery EEG data

C. Parameter setting and experimental environment

The model is trained in an environment with GPU acceleration. The total number of iteration rounds epoch is 3000, and the batch size is 32. The Adam optimizer is used. The initial learning rate is set to 0.001, the cross entropy loss function is used, and the training time of each round is recorded. The ten-fold cross validation method was used to verify the performance of the model in processing MI EEG signals, KFold=10.

TABLE III. EXPERIMENTAL ENVIRONMENT PARAMETERS TABLE

Experimental Environment	Parameter
CPU	Intel Core i5-10210U CPU @ 1.60GHz
GPU	NVIDIA GeForce RTX 2080 Ti
Memory	64 GB
Pytorch Version	1.8.1
Cuda Version	11.6
Python Version	3.7.0

D. Performance evaluation index

In this experiment, 10 fold cross validation was used to verify the EEG signals of each subject, and the model performance was evaluated by accuracy and Kappa value. Finally, the accuracy and Kappa values of each fold are averaged to obtain the final model performance evaluation parameters. The calculation is shown in Equation (7) (8).

$$Accuracy = (TP + TN) / (TP + TN + FP + FN) \quad (7)$$

$$Kappa = (p_0 - p_e) / (1 - p_e) \quad (8)$$

Where, TP represents the number of MI signals correctly identified by the model, TN represents the number of non-MI signals correctly identified by the model, FP represents the number of signals incorrectly marked as MI, and FN represents the number of MI signals that could not be detected. p_0 is the actual accuracy rate, and p_e is the expected consistency rate based on random guess, reflecting the expected accuracy rate if the model only relies on random selection.

E. Experimental process

1) Baseline method

In order to verify the performance of the proposed method in MI EEG signal recognition, four different baseline methods were used for comparative tests on two datasets, namely DRDA[32], SMM[33], DMDAN[34] and WNN[35]. Finally, the average Accuracy value obtained by the ten-fold cross validation method was used to measure the recognition performance of the proposed method for MI EEG signals.

Table IV shows the comparison results of the proposed method and the baseline method on data set 2a, and the data are from references. As can be seen from Table IV, the method

proposed in this paper has achieved a higher classification accuracy of MI EEG signals. For the three samples A02, A05 and A06 in dataset 2a, they are generally considered as representatives of large sample differences, which makes it difficult to classify these three samples. The method in this paper achieves an average recognition accuracy of 73.39%, far higher than the other four methods. Compared with the four baseline methods that all focus on feature extraction of EEG signals and ignore dynamic noise reduction based on the characteristics of EEG signals, this method combines the threshold learning sub network to conduct soft threshold noise reduction for signals on the one hand, and combines WPD to strengthen the focus on the ERS and ERD mesoscale of MI EEG signals, and ultimately can learn features with higher resolution, thus improving the accuracy of classification. At the same time, it can be seen from the table that the DRDA[32] model performs better than the method in this paper in subjects A05, A06 and A08, because the DRDA model takes into account the difference in the rhythmic range of EEG signals of different subjects or the same subject in different MI stages, and uses overlapping band filtering method to divide different sub bands according to the combination of different overlapping ranges, making the deep characteristics more obvious, but the final average accuracy and Kappa coefficient are lower than the method in this paper, that is, the stability of the model is not as good as the method in this paper.

TABLE IV. COMPARISON RESULTS OF AVERAGE CLASSIFICATION ACCURACY AND KAPPA COEFFICIENT BETWEEN THE PROPOSED METHOD AND THE BASELINE METHOD ON DATASET 2A

Model	Classification accuracy of different subjects (%)									Average accuracy(%)	Kappa coefficient
	A01	A02	A03	A04	A05	A06	A07	A08	A09		
DRDA ^[32]	83.19	55.14	77.43	75.28	72.29	67.15	76.18	93.61	72.00	74.70	0.6633
SMM ^[33]	81.94	59.38	81.60	62.85	59.03	49.36	86.11	77.78	78.47	70.72	0.6100
DMDAN ^[34]	84.72	61.11	92.36	69.44	62.85	57.64	88.89	85.42	86.11	76.50	0.6867
WNN ^[35]	82.86	68.21	87.86	72.50	61.79	57.86	87.50	89.64	76.07	76.03	0.6604
Proposed Method	93.75	81.74	89.68	79.36	71.45	66.98	91.26	92.36	93.47	84.45	0.8102

In order to show the classification results of this method on different categories, Figure 10 shows the average confusion matrix of 9 subjects on dataset 2a. From the results in Figure 10, we can see that the method in this paper has achieved more than 70% recognition accuracy in the four categories of left hand, right hand, feet and tongue, of which the recognition accuracy of left hand and feet is more than 80%, and the features extracted between different motion imagination categories are quite different. Among them, there is a small probability of confusion between the feet and the tongue, while the probability of confusion between the left hand and the right hand is low. It can be seen that the recognition performance of the method in this paper is higher than that of the other two types of movements.

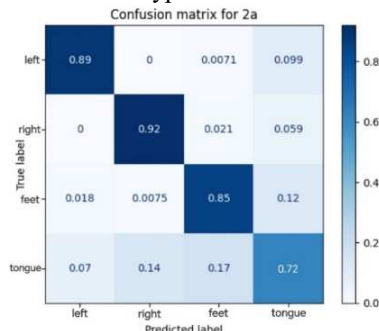


Fig. 10. The average confusion matrix of dataset 2a

Table V shows the comparison results of this method and the baseline method on data set 2b, and the data are from references. It can be seen from Table V that the method in this paper achieves a higher recognition accuracy of MI EEG signals, which is 4.94 percentage points higher than that of WNN. For B02 and B03 subjects with large difference in sample distribution, the recognition accuracy of this method also exceeds 80%. In addition, compared with the recent DMDAN, the recognition accuracy of this method in B04 to B08 subjects is slightly lower, which may be because the DRSN used in the experiment has a large number of stacking [3,4,6,3] of residual shrinkage blocks, which is not applicable to the two classification problem of data set 2b, so that excessive soft thresholding noise reduction leads to too many features being filtered, that is, the fineness of feature extraction may be too high, and the redundancy of feature information causes information loss. In addition, the mismatch between the complexity of the network design and the relative simplicity of the experimental task may also be one of the reasons, which leads to excessive noise reduction in specific subjects and weakens the retention of effective signal characteristics. But for the overall recognition accuracy and kappa value, the method in this paper has advantages.

TABLE V. COMPARISON RESULTS OF AVERAGE CLASSIFICATION ACCURACY AND KAPPA COEFFICIENT BETWEEN THE PROPOSED METHOD AND THE BASELINE METHOD ON DATASET 2B

Model	Classification accuracy of different subjects (%)									Average Accuracy(%)	Kappa coefficient
	B01	B02	B03	B04	B05	B06	B07	B08	B09		
DRDA ^[32]	65.00	56.79	54.06	95.63	74.69	79.06	80.00	87.81	82.81	75.09	0.5018
SMM ^[33]	67.81	51.79	53.44	93.31	82.81	74.69	72.19	82.50	75.62	72.68	0.4540
DMDAN ^[34]	76.56	73.57	82.19	95.94	98.12	83.13	91.56	95.00	82.50	86.50	0.7301
WNN ^[35]	86.86	70.63	85.72	87.69	81.93	75.68	86.45	91.35	77.36	82.63	0.7124
Proposed Method	91.68	85.96	84.36	92.36	81.45	80.36	88.43	93.71	89.81	87.57	0.8523

2) Ablation experiment

There are two key modules in this method, which are soft thresholding noise reduction module in DRSN and feature extraction module in WPD. In order to prove the effectiveness and necessity of the method proposed in this study, the effects of these two modules on the recognition accuracy were observed through ablation experiments. A total of three ablation experiments were designed: using only WPD feature extraction followed by baseline SMM classification (Experiment 1), using only DRSN for classification (Experiment 2), and using WPD for feature extraction and DRSN for classification and recognition (this method). Figure 11 shows the statistics of MI EEG classification accuracy of the three experiments on each subject in dataset 2a and 2b.

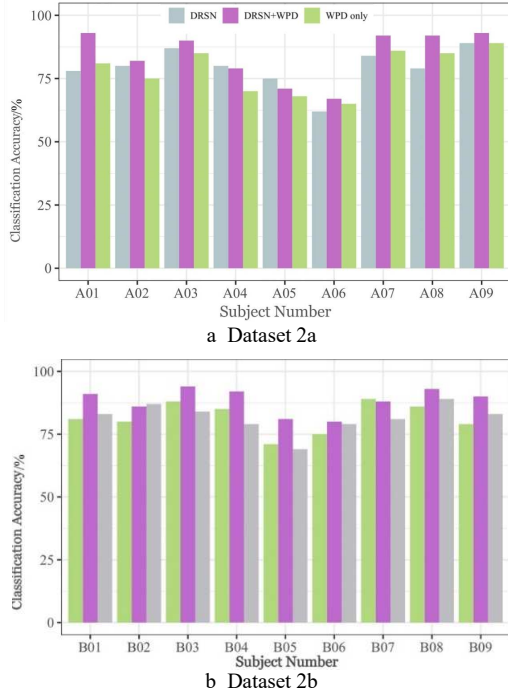


Fig. 11. Classification accuracy of different ablation experiment settings on datasets 2a and 2b for each subject

From the statistical results in Figure 11 (a), we can see that the method in this paper achieves the best classification recognition accuracy on subjects A01, A02, A03, A06, A07, A08, A09, experiment 1 achieves the best recognition accuracy on subjects A09, experiment 2 achieves the best recognition accuracy on subjects A03, A09. From the statistical results in Figure 11 (b), we can see that the method in this paper achieves the best classification accuracy on subjects B01, B03, B04, B05, B06, B08, B09, experiment 1 achieves the best classification accuracy on subjects B03, B07, and experiment 2 achieves the best classification accuracy on subject B08. The above results show that the feature extraction and noise reduction module combined with

WPD and DRSN can significantly improve the accuracy of EEG signal classification, especially for subjects with large differences in sample distribution. Specifically, the classification accuracy of this method on most subjects is higher than that of DRSN or WPD alone, which indicates that the combination of multi-scale features extracted by WPD and DRSN can effectively capture the ERD and ERS intermediate scale features in EEG signals, avoid too many features being filtered out, and improve the classification performance. The same noise reduction threshold is not applicable to different EEG signals because of the nonlinearity and noise specificity of MI EEG signals. However, this method considers the learning ability of DRSN for different noise reduction thresholds and the use of WPD to enhance DRSN's ability to capture ERD and ERS midscale features, so it can obtain the best classification and recognition accuracy of MI EEG signals on datasets 2a and 2b.

TABLE VI. AVERAGE CLASSIFICATION ACCURACY OF DIFFERENT ABLATION EXPERIMENT SETTINGS ON TWO DATASETS

Model	Dataset2a(%)	Dataset2b(%)
Experiment 1	70.19	75.42
Experiment 2	75.36	77.21
Proposed Method (DRSN+WPD)	84.45	87.57

According to the results in Table VI, compared with ablation experiments 1 and 2, this method has achieved higher average classification accuracy on both datasets. Compared with experiment 1 and this method, the introduction of DRSN improved the classification accuracy by 14.26 and 12.15 percentage points on dataset 2a and 2b, respectively; Compared with experiment 2 and the method in this paper, WPD was introduced on the basis of DRSN, and the classification accuracy on the two datasets increased by 9.09 and 10.36 percentage points respectively. The above results show that DRSN and WPD used in this paper can improve the classification accuracy of MI EEG signals, and the combination of the two can further improve the classification accuracy.

3) Experiment on the Optimal Stack Number of BasicBlock Modules

DRSN is composed of multiple BasicBlock modules stacked in layers. Therefore, the number of BasicBlock modules stacked is one of the factors that affect the performance of the model in EEG signal recognition, which directly determines the depth and complexity of the model. In this experiment, on data sets 2a and 2b, on the basis of DRSN combined with WPD, the number of different residual blocks is used to record the model training time and evaluation indicators, and the results are compared with the corresponding parameters of the network without WPD

fusion, so as to find the optimal number of BasicBlock modules stacked by the model for MI EEG signal recognition.

The specific results are shown in Table VII.

TABLE VII. COMPARISON OF THE IMPACT OF DIFFERENT STACKING NUMBERS OF BASICBLOCK MODULES ON THE PERFORMANCE OF DRSN MODELS

Model	Number of BasicBlock stacks	Dataset 2a			Dataset 2b		
		training time	Accuracy(%)	Kappa(%)	training time	Accuracy(%)	Kappa(%)
DRSN	[2,2,2,2]	9634s	75.24	66.97	5523s	86.41	79.35
	[3,4,6,3]	13585s	73.09	64.10	9134s	81.34	76.28
	[4,6,8,4]	18494s	70.64	60.87	14872s	76.89	76.98
DRSN + WPD	[2,2,2,2]	983s	63.10	51.00	569s	91.26	89.64
	[3,4,6,3]	1395s	84.45	81.02	998s	87.57	85.23
	[4,6,8,4]	1801s	71.59	62.38	1523s	79.53	76.72

It can be seen from the data in Table VII that in the DRSN model, the number of different BasicBlock stacks has a significant impact on the performance of DRSN in recognizing MI EEG signals.

The experiment uses [2,2,2,2] as the initial stack number configuration, aiming to establish a benchmark model to evaluate the impact of deeper network structure on performance. This configuration ensures that the depth of the network is sufficient to learn complex features, while avoiding the over fitting problem that may be caused by excessive complexity. It means that a total of four residual shrink blocks are used to form the DRSN. Each block is composed of two BasicBlock modules. This configuration uses the network without WPD fusion in dataset 2a and shows the best performance in dataset 2b.

Then, try the symmetric stacking configuration of [3,4,6,3]. The purpose of this configuration is to focus on the extraction of intermediate scale features. By focusing the model's attention on the middle layer, it can more effectively capture the intermediate scale features of ERS and ERD generated by alpha and beta band waves subdivided by WPD, and also enable the model to better reconstruct signals while extracting features. This configuration performs well in the network after WPD fusion is used in dataset 2a.

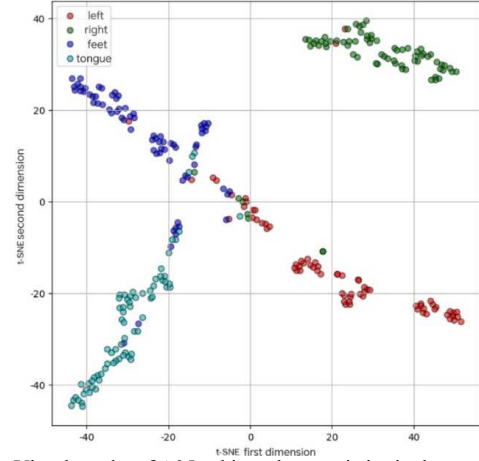
Finally, try to use the number of BasicBlock stacks of [4,6,8,4] for experiments. At this time, the recognition performance of both networks has declined, which means that after a certain point, increasing the network depth will no longer bring about performance improvement. At this time, if the network is too deep, over fitting may occur.

It can also be seen from the table that the training time of DRSN+WPD is significantly reduced compared with the network without WPD fusion. Under the same stack number, the training time of DRSN with WPD fusion is nearly nine times less than that of the network without WPD fusion. Therefore, in the recognition of MI EEG signals, it is very important to select the appropriate DRSN network depth to improve the accuracy of network recognition. At the same time, WPD, as a means of fine decomposition of EEG signals, can improve the performance of DRSN and accelerate model convergence, but it is also necessary to find a suitable network depth to maximize its effect. The experimental results show that the [3,4,6,3] configuration of the BasicBlock module in this method achieves the best balance in dataset 2a, and the [2,2,2,2] configuration achieves the best balance in dataset 2b.

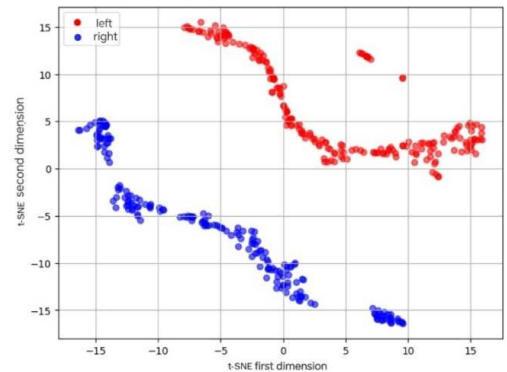
4) Feature visualization

In order to visualize the features extracted by the method

in this paper, A05 and B05 subject samples were selected from data set 2a and 2b, respectively, to show different categories of features learned by the model. In the experiment, the number of BasicBlock module stacks of [3,4,6,3] and [2,2,2,2] were used for two test samples, and the DRSN model was trained for 300 iterations to obtain the feature vectors output by the model, and the t-SNE tool was used to reduce the features to two-dimensional display. Figure 12 shows the extracted feature visualization results on A05 and B05 test sets respectively.



a Visual results of A05 subject characteristics in dataset 2a



b Visual results of B05 subject characteristics in dataset 2b

Fig. 12. Visualization of Subject Characteristics in Different Dataset

As can be seen from Figure 12, for the two test samples, after 300 rounds of iterative training, the DRSN model can obtain features with strong resolution, and there are obvious boundaries between samples of different categories, especially for the data set 2b, the classification effect is very good. The features of both left and right hand categories are highly clustered within the class, and the features of between classes are far clustered. Therefore, this method can achieve higher accuracy of MI EEG signal classification and

recognition. Among them, there are still some samples that cannot be separated between the categories in Figure 12 (a). In fact, A05 subjects in data set 2a contain more noise, which is more difficult to classify than other subjects. However, the classification of A05 subjects in this method still maintains the characteristics of high clustering within the class, and the network architecture needs to be further optimized in the future, in order to obtain higher recognition accuracy in the difficult to distinguish MI class.

IV. CONCLUSION

Faced with the problem of noise reduction and difficulty in classification and recognition of MI EEG signals, this paper proposes the fusion of WPD and DRSN for noise reduction and feature extraction of MI EEG signals. The proposed method enhances the ability of DRSN model to capture the mid-scale features of MI EEG signals. Through WPD, the signal is decomposed into different frequency bands, including α and β bands most related to MI. The model realizes more precise time-frequency feature recognition, thus improving the accuracy and efficiency of classification. The experimental results on two representative data sets 2a and 2b confirm that the proposed fusion method in this paper has significantly improved the performance compared with the traditional methods, and has achieved an average classification accuracy of 84.45% and 91.26% respectively on the two data sets. Future research can explore more effective feature extraction techniques and network architecture simplification methods to further improve the classification performance of DRSN for samples with different classification complexity. At the same time, the new method can be extended to other MI EEG signal classification and recognition applications.

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